

Technology Stack

Date	26 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Bootstrap	Provides a simple, responsive, and user-friendly interface for entering soil and environmental parameters and displaying crop recommendations
2	Backend / Server-Side	Python, Flask, Scikit-learn	Processes user inputs, loads the trained machine learning model, and predicts the most suitable crop based on soil and climatic conditions. .
3	Database / Data Storage	CSV Dataset, Pickle (.pkl)	Stores the crop dataset used for training and the serialized machine learning model used during prediction.
4	Deployment	Flask Local Server	Hosts the OptiCrop web application and enables real-time crop recommendation through a browser.
5	Version Control & CI/CD	Git, GitHub	Maintains project versions, supports team collaboration, and tracks code changes. .
6	Third-Party APIs / Other Tools	Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Pickle	Used for data preprocessing, exploratory data analysis (EDA), visualization, model training, evaluation, and saving/loading the trained model.